

Exercise JA

Q1. For $x \in (0, \pi)$ the equation $\sin x + 2\sin 2x - \sin 3x = 3$ has

- (A) infinitely many solutions (B) three solutions (C) one solution
(D) no solution

Ans. D

Sol. -

Q2. The number of distinct solutions of the equation $\frac{5}{4} \cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2$ in the interval $[0, 2\pi]$ is

Ans. 8

Sol.
$$\frac{5}{4} \cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2$$
$$\Rightarrow \frac{5}{4} \cos^2 2x + 1 - \frac{1}{2} \sin^2 2x + 1 - \frac{3}{4} \sin^2 2x = 2$$

$$\Rightarrow \cos^2 2x = \sin^2 2x$$

$$\Rightarrow \tan^2 2x = 1$$

$$2x \in [0, 4\pi] \Rightarrow x = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}, \frac{13\pi}{8}, \frac{15\pi}{8}$$

so number of solution is 8

Q3. Let $S = \left\{ x \in (-\pi, \pi) : x \neq 0, \pm \frac{\pi}{2} \right\}$. The sum of all distinct solutions of the equation $\sqrt{3} \sec x + \operatorname{cosec} x + 2(\tan x - \cot x) = 0$ in the set S is equal to

- (A) $-\frac{7\pi}{9}$ (B) $-\frac{2\pi}{9}$ (C) 0 (D) $\frac{5\pi}{9}$

Ans. C

Sol.
$$2 \left(\frac{\sqrt{3}/2}{\cos x} + \frac{1/2}{\sin x} \right) + 2(\tan x - \cot x) = 0$$

$$\Rightarrow 2 \frac{\cos(x - 60^\circ)}{\sin x \cos x} + \frac{2(-\cos 2x)}{\sin x \cos x} = 0$$

$$\Rightarrow \cos(x - 60^\circ) = \cos 2x$$

$$\Rightarrow x = 2n\pi - \frac{\pi}{3}, (6m + 1)\frac{\pi}{9} (m, n \in I)$$

$$\Rightarrow \text{Solutions belonging to set } S = -\frac{\pi}{3}, -\frac{5\pi}{9}, \frac{\pi}{9}, \frac{7\pi}{9}$$

sum = 0

Q4. Let a, b, c be three non-zero real numbers such that the equation $\sqrt{3}a \cos x + 2b \sin x = c$ $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, has two distinct real roots α and β with $\alpha + \beta = \pi/3$. Then, the value of b/a is _____ .

Ans. 0.5

Sol. -

Q5. Consider the following lists :

List - I	List - I
(I) $\left\{x \in \left[-\frac{2\pi}{3}, \frac{2\pi}{3}\right] : \cos x + \sin x = 1\right\}$	(P) has two elements
(II) $\left\{x \in \left[-\frac{5\pi}{18}, \frac{5\pi}{18}\right] : \sqrt{3} \tan 3x = 1\right\}$	(Q) has three elements
(III) $\left\{x \in \left[-\frac{6\pi}{5}, \frac{6\pi}{5}\right] : 2 \cos(2x) = \sqrt{3}\right\}$	(R) has four elements
(IV) $\left\{x \in \left[-\frac{7\pi}{4}, \frac{7\pi}{4}\right] : \sin x - \cos x = 1\right\}$	(S) has five elements
	(T) has six elements

The correct option is:

- (A) (I) $\rightarrow$ (P); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (S)
 (B) (I) $\rightarrow$ (P); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (R)
 (C) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (S)
 (D) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (R)

Ans. 6313255cf69e6e3404aecd3e

Sol. $\left\{x \in \left[-\frac{2\pi}{3}, \frac{2\pi}{3}\right], \cos x + \sin x = 1\right\}$

$$\cos x + \sin x = 1$$

$$\sin\left(\frac{\pi}{4} + x\right) = \frac{1}{\sqrt{2}}$$

$$\frac{\pi}{4} + x = n\pi + (-1)^n \frac{\pi}{4}$$

$$x = n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{4}$$

$\therefore$ has 2 elements. $\rightarrow$ P

$$(ii) \left\{x \in \left[-\frac{5\pi}{18}, \frac{5\pi}{18}\right] : \sqrt{3} \tan 3x = 1\right\}$$

$$\sqrt{3} \tan 3x = 1$$

$$\tan 3x = \frac{1}{\sqrt{3}}$$

$$3x = n\pi + \frac{\pi}{6}$$

$$x = \frac{n\pi}{3} + \frac{\pi}{18}$$

$\therefore$ has 2 elements. $\rightarrow$ P

$$(iii) \left\{x \in \left[-\frac{6\pi}{5}, \frac{6\pi}{5}\right] : 2 \cos 2x = \sqrt{3}\right\}$$

$$2 \cos 2x = \sqrt{3}$$

$$\cos 2x = \frac{\sqrt{3}}{2}$$

$$2x = 2n\pi \pm \frac{\pi}{6}$$

$$x = n\pi \pm \frac{\pi}{12}$$

$\therefore$ has 6 elements. $\rightarrow T$

$$(iv) \left\{ x \in \left[-\frac{7\pi}{4}, \frac{7\pi}{4} \right] : \sin x - \cos x = 1 \right\}$$

$$\sin x - \cos x = 1$$

$$\sin \left(x - \frac{\pi}{4} \right) = \frac{1}{\sqrt{2}}$$

$$x - \frac{\pi}{4} = n\pi + (-1)^n \frac{\pi}{4}$$

$$x = n\pi + (-1)^n \frac{\pi}{4} + \frac{\pi}{4}$$

$\therefore x$ has 4 elements. $\rightarrow R$

$\therefore$ option B is correct